



KOLEJ MATRIKULASI NEGERI SEMBILAN
UJIAN SELARAS 2
SESI 2024/2025
DP014 FIZIK (2 JAM)

NAMA : _____

PRAKTIKUM : _____

JANGAN BUKA KERTAS SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.

<i>Untuk Kegunaan Pemeriksa</i>		
NO. SOALAN	MARKAH PENUH	MARKAH DIPEROLEH
1	8	
2	12	
3	9	
4	12	
5	9	
6	12	
7	18	
JUMLAH	80	

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Speed of light in vacuum <i>Laju cahaya dalam vakum</i>	c	$= 3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space <i>Ketelapan ruang bebas</i>	μ_o	$= 4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space <i>Ketelusan ruang bebas</i>	ε_o	$= 8.85 \times 10^{-12} \text{ F m}^{-1}$
Electron charge magnitude <i>Magnitud cas elektron</i>	e	$= 1.60 \times 10^{-19} \text{ C}$
Planck constant <i>Pemalar Planck</i>	h	$= 6.63 \times 10^{-34} \text{ J s}$
Electron mass <i>Jisim elektron</i>	m_e	$= 9.11 \times 10^{-31} \text{ kg}$ $= 5.49 \times 10^{-4} \text{ u}$
Neutron mass <i>Jisim neutron</i>	m_n	$= 1.674 \times 10^{-27} \text{ kg}$ $= 1.008665 \text{ u}$
Proton mass <i>Jisim proton</i>	m_p	$= 1.672 \times 10^{-27} \text{ kg}$ $= 1.007277 \text{ u}$
Deuteron mass <i>Jisim deuteron</i>	m_d	$= 3.34 \times 10^{-27} \text{ kg}$ $= 2.014102 \text{ u}$
Molar gas constant <i>Pemalar gas molar</i>	R	$= 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Rydberg constant <i>Pemalar Rydberg</i>	R_H	$= 1.097 \times 10^7 \text{ m}^{-1}$
Avogadro constant <i>Pemalar Avogadro</i>	N_A	$= 6.02 \times 10^{23} \text{ mol}^{-1}$
Boltzmann constant <i>Pemalar Boltzmann</i>	k	$= 1.38 \times 10^{-23} \text{ J K}^{-1}$
Gravitational constant <i>Pemalar graviti</i>	G	$= 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Free-fall acceleration <i>Pecutan jatuh bebas</i>	g	$= 9.81 \text{ m s}^{-2}$

LIST OF SELECTED CONSTANT VALUES
SENARAI NILAI PEMALAR TERPILIH

Atomic mass unit <i>Unit jisim atom</i>	1 u	$= 1.66 \times 10^{-27} \text{ kg}$ $= 931.5 \frac{\text{MeV}}{c^2}$
Electron volt <i>Elektron volt</i>	1 eV	$= 1.6 \times 10^{-19} \text{ J}$
Constant of proportionality for Coulomb's law <i>Pemalar hukum Coulomb</i>	$k = \frac{1}{4\pi\epsilon_0}$	$= 9.0 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
Atmospheric pressure <i>Tekanan atmosfera</i>	1 atm	$= 1.013 \times 10^5 \text{ Pa}$
Density of water <i>Ketumpatan air</i>	ρ_w	$= 1000 \text{ kg m}^{-3}$

LIST OF SELECTED FORMULAE
SENARAI RUMUS TERPILIH

- | | |
|--|---|
| 1. $v = u + at$ | 20. $v = r\omega$ |
| 2. $s = ut + \frac{1}{2}at^2$ | 21. $a_t = r\alpha$ |
| 3. $v^2 = u^2 + 2as$ | 22. $\omega = \omega_o + \alpha t$ |
| 4. $s = \frac{1}{2}(u + v)t$ | 23. $\theta = \omega_o t + \frac{1}{2}\alpha t^2$ |
| 5. $v = u - gt$ | 24. $\omega^2 = \omega_o^2 + 2\alpha\theta$ |
| 6. $v^2 = u^2 - 2gs$ | 25. $\theta = \frac{1}{2}(\omega_o + \omega)t$ |
| 7. $s = ut - \frac{1}{2}gt^2$ | 26. $\tau = rF \sin \theta$ |
| 8. $p = mv$ | 27. $\frac{Q}{t} = -kA \left(\frac{\Delta T}{L} \right)$ |
| 9. $J = F\Delta t$ | 28. $v_{rms} = \sqrt{\langle v^2 \rangle}$ |
| 10. $J = \Delta p = mv - mu$ | 29. $v_{rms} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$ |
| 11. $f_s \leq \mu_s N$ | 30. $\Delta U = Q - W$ |
| 12. $f_k = \mu_k N$ | |
| 13. $W = \vec{F} \cdot \vec{s} = Fs \cos \theta$ | |
| 14. $K = \frac{1}{2}mv^2$ | |
| 15. $U = mgh$ | |
| 16. $U_s = \frac{1}{2}kx^2 = \frac{1}{2}Fx$ | |
| 17. $a_c = \frac{v^2}{r} = r\omega^2 = v\omega$ | |
| 18. $F_c = \frac{mv^2}{r} = mr\omega^2 = mv\omega$ | |
| 19. $s = r\theta$ | |

Answer **all** questions.

- 1 (a) Light takes 8.2 minutes to travel from Sun to Earth. If it travels at constant speed $300,000 \text{ km s}^{-1}$, calculate the distance between Sun and Earth in SI unit. [2 marks]

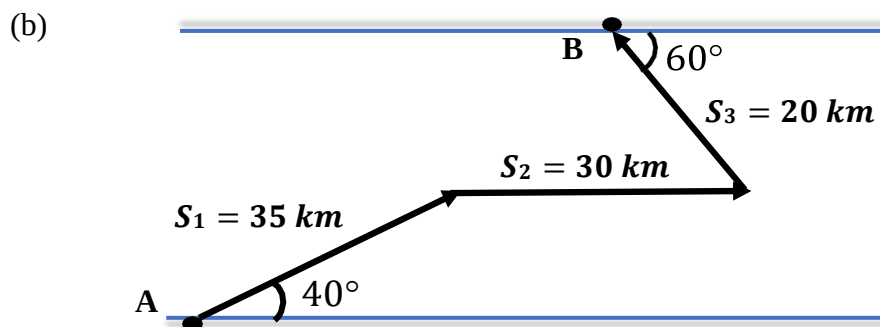


FIGURE 1

A school bus travels from Point A to Point B with different displacement as shown in **FIGURE 1**. Calculate the magnitude and direction of the resultant displacement followed by the school bus.

[6 marks]

- 2 (a) A car travelling at 91.0 km h^{-1} approaches a red traffic light. If the driver applies the brake with a deceleration of 6.40 m s^{-2} , calculate the distance of the car until it stops.

[3 marks]

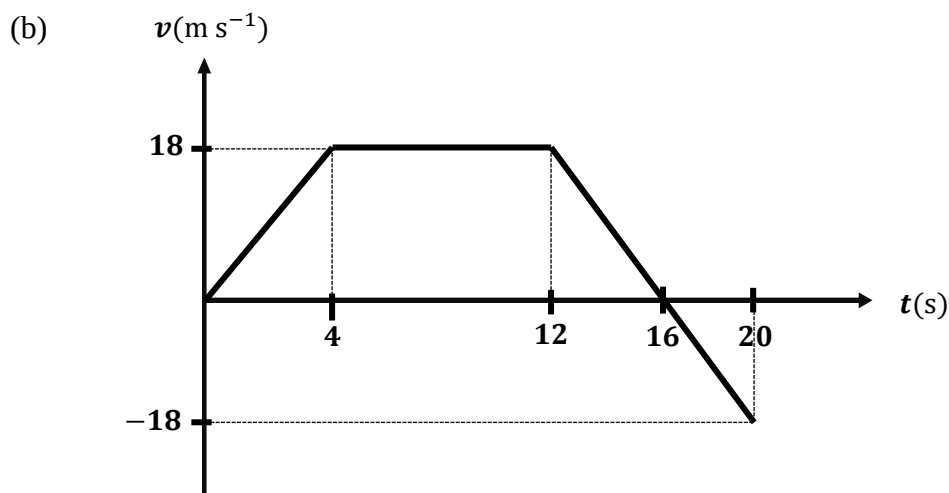


FIGURE 2

FIGURE 2 shows the velocity-time graph of the motion for a moving object in a straight line. Calculate the

- displacement for the whole journey.
- acceleration of the object at $t = 16 \text{ s}$.

[5 marks]

- (c) A tennis ball is thrown vertically upward from the ground with an initial speed, u . The tennis ball hits the ground 5 s later. Calculate the
- initial speed, u of the tennis ball.
 - maximum height reached by the tennis ball.

[4 marks]

3 (a)

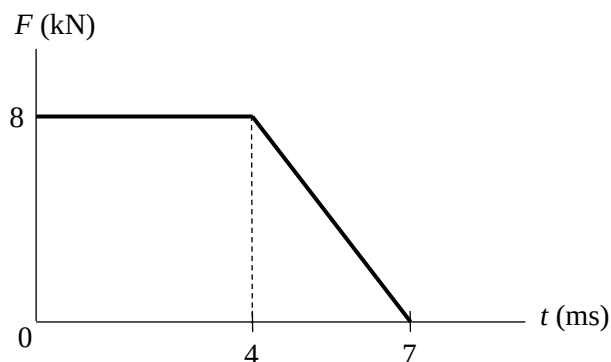


FIGURE 3.1

FIGURE 3.1 shows the force exerted on an object of mass 1.5 kg for 7 ms. Calculate the total impulse of the object.

[3 marks]

(b)

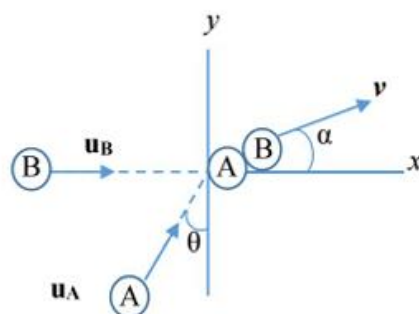


FIGURE 3.2

FIGURE 3.2 shows two identical balls colliding obliquely and sticking together after collision. Calculate the magnitude and direction of velocity after the impact.

(Given $u_A = u_B = 55 \text{ m s}^{-1}$ and $\theta = 20^\circ$)

[6 marks]

4 (a)

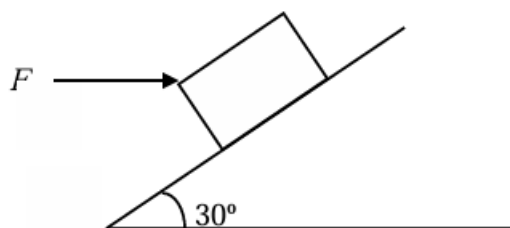


FIGURE 4.1

FIGURE 4.1 shows an object of mass 2.0 kg placed on a rough plane inclined at 30° with the horizontal. The coefficient of kinetic friction between the object and the plane surface is 0.25. A constant horizontal force, $F = 50$ N acts on the object and pushes it along the inclined plane with acceleration, a .

- (i) Sketch a free body diagram showing all the forces acting on the object.
- (ii) Calculate the acceleration of the object.

[8 marks]

(b)

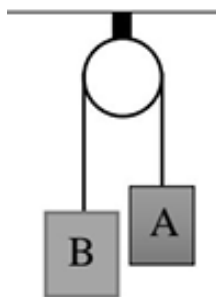
**FIGURE 4.2**

FIGURE 4.2 shows two boxes **A** and **B** hung by a light string through a frictionless pulley. The mass of box **B** is 0.50 kg and the mass of box **A** is 0.25 kg. Calculate the tension in the string.

[4 marks]

- 5 (a) A stone at the end of a string of length 1.2 m is whirled in a horizontal circle on a smooth surface. The speed of the stone is 0.75 m s^{-1} . Calculate the

- (i) angular velocity of the stone.
- (ii) number of revolutions made by the stone make in 2 minutes.

[3 marks]

- (b) An object of mass 100 g is placed on a horizontal turn-table. If the maximum frictional force between the object and turn-table is 0.55 N,

- (i) sketch the free body diagram to show all the forces acting on the object.
- (ii) calculate the centripetal acceleration of the object.

[6 marks]

- 6 (a) The angular velocity of a wheel decreases from $20\pi \text{ rad s}^{-1}$ to $10\pi \text{ rad s}^{-1}$ in 6.0 s. Calculate the

- (i) angular acceleration.
- (ii) number of rotations made by the wheel.
- (iii) extra time needed to stop the wheel.

[7 marks]

(b)

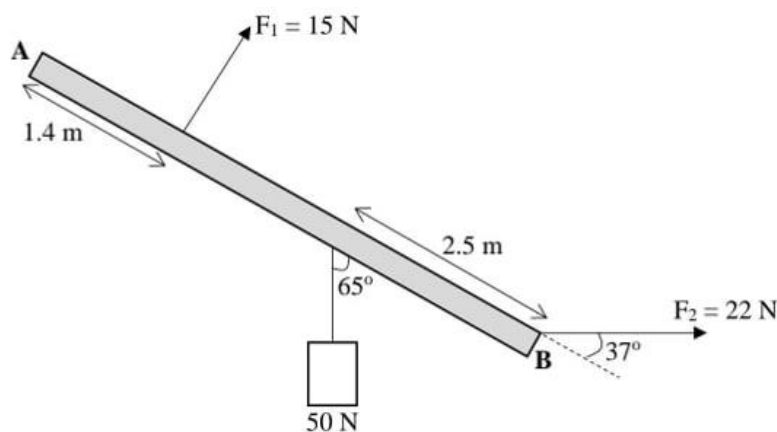
**FIGURE 6**

FIGURE 6 shows two forces F_1 and F_2 of 15 N and 22 N respectively exerted on a uniform plank of 6 m. A hanging load having weight of 50 N is hung 2.5 m from point B. Calculate the magnitude of

- (i) total torques at point **A** (Assume the plank is massless).
- (ii) total torques at point **B** if the mass of the plank is 2 kg.

[5 marks]

- 7 (a) A glass window is 0.50 cm thick with an area of 0.323 m^2 . The inside temperature is 30°C and outside temperature is 48°C . (Given the thermal conductivity of glass is $0.84 \text{ W m}^{-1} \text{ K}^{-1}$)

- (i) Calculate the rate of heat flow through this window.
- (ii) Calculate the heat flow through this window in one minute.
- (iii) Sketch a graph of temperature against distance for heat conduction for insulated material.

[6 marks]

- (b) (i) Calculate the root mean square speed of an oxygen molecule in a sample of oxygen gas at 2°C ? (Given the molar mass of oxygen is 32 g mol^{-1})
- (ii) The rms speed of a hydrogen atom at a temperature of 25°C is 600 m s^{-1} . Calculate the mass of a hydrogen atom.

[5 marks]

- (c) (i) An ideal gas absorbs 1000 J of heat as it expands. Work done by the gas is 20 J. Calculate the change in the internal energy of the gas.
- (ii) A sample of gas expands isothermally and does $4.0 \times 10^3 \text{ J}$ of work in the process. Calculate the heat absorbed by the gas.

[7 marks]

END OF QUESTIONS PAPER
KERTAS SOALAN TAMAT